MorphMuse

Plugin Guide

Ricardo Y. Maeda – April 2026

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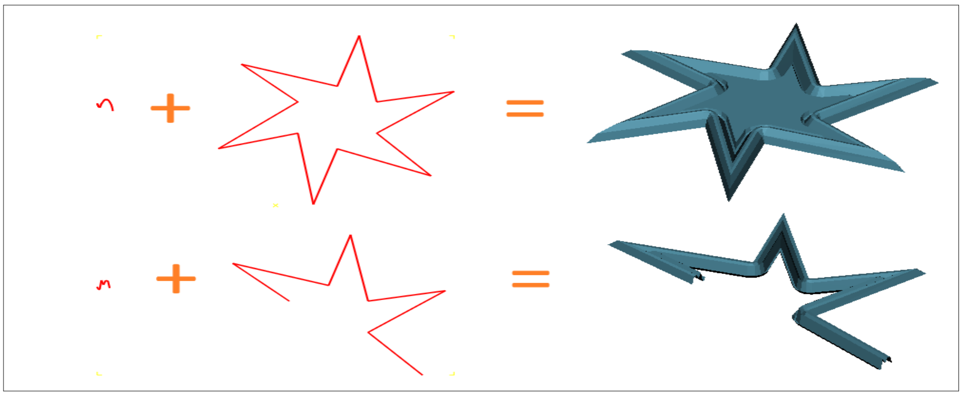
1. Overview

MorphMuse allows you to build surface geometries in CamBam by combining one open curve with one closed outline, or by combining two open curves. It automatically generates a smooth surface:

> One of the curves acts as the surface generatrix, with its starting point sliding along the other curve to generate the surface.



Installation

Before installation, choose the plugin file that matches your CamBam version:

- Available at: atelier-des-fougeres.fr/Cambam/Aide/Plugins/MorphMuse.html
- Two builds are available: 32-bit and 64-bit.
    - **morphmuse_x86.dll** → for 32-bit CamBam installations.
    - **morphmuse_x64.dll** → for 64-bit CamBam installations.

Installation steps:

1. Copy the appropriate `.dll` file into the `/plugins` directory of your CamBam installation.

2. Restart CamBam to load the plugin.
3. The plugin will appear in the CamBam menu.

Functionality

MorphMuse generates 3D surfaces by combining polylines:

- **Open + Closed Polyline** → creates a surface swept along the open curve, capped at the end.
- **Open + Open Polyline** → creates a surface swept along one open curve (rail) by another open curve (profile).

This enables machinists to build complex surfaces directly from 2D geometry.

Step-by-Step Usage

1. **Select Input Geometry**

    - Choose one open polyline and one closed polyline, OR two open polylines.

2. **Run the Plugin**

    - Access MorphMuse from the CamBam plugin menu.
    - If two open polylines are selected, a dialog will ask which one should act as the rail.

3. **View the Result**

    - The generated surface will appear in a new layer.

4. **Adjust Orientation if Needed**

    - The direction of the closed polyline (clockwise vs. counterclockwise) affects the sweep.
    - The starting point of the open polyline influences the surface.
    - If the result looks incorrect, try reversing the open polyline.

Adjustment

Internally, the plugin simplifies curves to generate surfaces with appropriate precision.

- To adjust precision, assign a new value to the **step resolution** parameter in: `[Tools] → [Options]`.

Limitations

- Works only with 2D polylines on the XY plane.
- Requires at least one closed and one open curve, OR two open curves.
- The undo feature is not fully functional.
- Surface generation may fail if the geometry is too complex.

Notes for Use

- Keep curves simple and clean to avoid errors.
- Always check orientation before running the plugin.
- Generated surfaces are best used as a starting point for machining paths.

Development Note

This plugin was initially developed with assistance from **Microsoft Copilot**, **Meta Manus**, and **GitHub Copilot**, which supported code generation, logic refinement, and documentation drafting. While AI assistants played a valuable role in accelerating development, all final decisions, testing, integration, and bug fixes were carried out manually.
This project is independently maintained and **not affiliated with or endorsed by Microsoft or Meta**.

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9. How to Simplify a Mesh in MeshLab

Mesh simplification reduces the number of faces or vertices in a mesh while preserving its overall shape. This is useful for optimizing performance, reducing file size, or preparing models for export.

Step-by-Step Instructions:

1. **Open the Mesh**
   - Go to *File → Import Mesh* and select your mesh file.
2. **Select the Mesh**
   - Ensure your mesh is selected in the *Layer Panel* on the right.
3. **Access the Simplification Filter**
   - *Filters → Remeshing, Simplification and Reconstruction → Quadric Edge Collapse Decimation*.
4. **Configure Parameters**
   - **Target number of faces**: set the desired number (e.g., 1000).
   - **Preserve Boundary**: keeps mesh edges intact.
   - **Preserve Normal**: maintains smooth shading and surface appearance.
   - **Quality Threshold**: leave at 1.0 for maximum detail retention.
5. **Apply the Filter**

    - Click *Apply* to simplify the mesh. The result will appear immediately in the viewport.

6. **Check the Result**

    - Go to *Render → Show Vertex/Face Count* to view updated counts.

7. **Export the Simplified Mesh**

    - *File → Export Mesh As...* and save your simplified model.